Star Plucking: A Monotone $n^{3-o(1)}$ Lower Bound for 3-List Orthogonal Vectors

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Work in progress: not verified

This report records work in progress; it is not peer-reviewed.

- **AI-generated:** proofs, computations and code were produced by Claude (Anthropic, Claude Opus 5.5) in sessions directed by the author, who has not independently verified the mathematics.
- **No human review:** the main proof was checked only by AI referee agents of the same model family (`docs/ai-review-log.md`).
- **No formal verification:** nothing is formalized in Lean; SAT results have no DRAT certificates.

Everything here still needs thorough verification. Do not cite these results as established.

Abstract

We show that every monotone circuit deciding whether three lists of n Boolean vectors of dimension $d \geq c(\varepsilon) \log n$ have no orthogonal triple needs $n^{3-\varepsilon}$ gates. This extends the recent two-list theorem of Choudhury, Limaye, Sreenivasaiah and Srinivasan [3] to three lists at logarithmic dimension, a case they leave open. The new ingredient, *star plucking*, controls approximator conjuncts that share a core vector. A corollary is an $n^{3-o(1)}$ monotone bound for unbalanced OV (n versus n^2 vectors). We also give three results on routes to refuting the Orthogonal Vectors Hypothesis (OVH) after the refutation of ETH [1, 2]: a collapse for block-CNF, a no-go for linear deformation designs, and exact equality ranks of DISJ_k for $k \leq 4$. None of this proves or disproves OVH.

1 Introduction

The Orthogonal Vectors problem $\mathrm{OV}_{n,d}$ asks whether two lists of n vectors in $\{0,1\}^d$ contain an orthogonal pair. The *Orthogonal Vectors Hypothesis* (OVH) states that for every $\varepsilon > 0$ there is c such that $\mathrm{OV}_{n,c\log n}$ has no $O(n^{2-\varepsilon})$ algorithm. Its k-list version (k-OVH) asks for $n^{k-\varepsilon}$. OVH was mainly supported by SETH, which implies it [4]. The October 2026 $2^{o(n)}$ algorithms for 3-SAT [1, 2] refute ETH and SETH but not OVH, since their running times degrade with clause width. OVH is thus open and without its main evidence, which raises two questions: which routes could refute it, and in which restricted models can it be proved?

Contributions.

1. **Main result (§2).** Monotone circuits for $3\mathrm{Int}_{n,d} = \neg\, 3\text{-OV}$ need $n^{3-\varepsilon}$ gates at $d = O(\varepsilon^{-3}) \log n$ (Theorem 1). Previously this was known only for two lists [3, Thm 5], and for $k \geq 3$ lists only at $d = n^{\Omega(1)}$ [3, Thm 4]. Corollary 2 transfers the bound to unbalanced two-list OV.

2. **Limits of refutation routes (§3):**
   - a collapse theorem for block-CNF (Theorem 8);
   - a no-go for linear deformation designs (Proposition 9);
   - exact equality ranks with an LP barrier (Theorem 10, Propositions 11 and 12).

Scope. Monotone lower bounds do not transfer to general algorithms [8, 10]. The main theorem only shows that refuting 3-list OVH requires non-monotone steps such as cancellation.

2 The main result

For lists U, V, Z of n vectors in $\{0,1\}^d$, let $3\mathrm{Int}_{n,d}(U,V,Z) = 1$ iff $u \wedge v \wedge z \neq 0^d$ for all $(u,v,z) \in U \times V \times Z$. This function is monotone.

Theorem 1 (main)**.** *For every $\varepsilon \in (0,1)$ there is $c = O(\varepsilon^{-3})$ such that, for all large n and all $d \geq c \log n$, every monotone circuit computing $3\mathrm{Int}_{n,d}$ has at least $n^{3-\varepsilon}$ gates.*

Corollary 2 (unbalanced OV)**.** *Let $\mathrm{Int}_{n,m,d}$ be the negation of OV with lists of sizes n and m. Under the hypotheses of Theorem 1, every monotone circuit for $\mathrm{Int}_{n,n^2,d}$ has at least $n^{3-\varepsilon} - n^2 d$ gates.*

Proof. $3\mathrm{Int}(U,V,Z) = \mathrm{Int}(U,W)$ with $W = \{v \wedge z\}$, whose n^2 vectors take $n^2 d$ monotone AND gates to compute. □

2.1 Overview

We use Razborov's approximation method [8, 9] in the form of [3]. Each gate is replaced by an *approximator*: a conjunction of monotone pieces, each reading a few vectors. Errors are measured on a distribution $\mathcal{D}_0$ of 0-inputs and a distribution $\mathcal{D}_1$ of 1-inputs. For two lists, CLSS use pieces that each read one vector. For three lists, one-vector pieces lose a factor of only n^{-2} at OR gates, which caps the bound at about n^2, so pieces must read two vectors. This creates a new failure mode, *stars*: many pieces $B_i(u, v_i)$ sharing a core u. When two stars with the same core meet at an OR gate, $\Theta(n^2)$ three-vector pieces must be dropped, at a $\mathcal{D}_0$-cost of $\Theta(1/n)$ per gate. *Star plucking* (conditions (A3)–(A4)) fixes this.

- For each value w of the core, petals that rarely fail under $\mathcal{D}_1$ are switched off. This is cheap by Lemma 4.
- A one-vector check $Q_u(w)$, "few frequently-failing petals", is added. This is cheap on $\mathcal{D}_1$, because many such petals make the star almost surely 0 anyway.

After plucking, the petal that matters on a $\mathcal{D}_0$ sample is a bad one only with probability $\approx 1/n$. That restores an $n^{-3+o(1)}$ error per gate.

2.2 Distributions

$\mathcal{D}_1 = \mu_p$, i.e. each bit is 1 independently with probability p. By a union bound, $\Pr_{\mathcal{D}_1}[3\mathrm{Int} = 0] \leq n^3(1-p^3)^d \leq \frac{1}{2}$ once $d \geq (3 \ln n + 1)/p^3$.

$\mathcal{D}_0$: pick $a, b, c \in [n]$ uniformly and call u_a, v_b, z_c *chosen*. For each coordinate t independently, pick $L_t \in \{U, V, Z\}$ uniformly. The chosen vector of list L_t gets 0 at t, the other two chosen vectors get 1, and all unchosen vectors are 1^d. Every sample is a 0-input.

Lemma 3. *(i) The indices are independent of the contents.*

(ii) Given the indices, each chosen vector has law $\mu_{2/3}$.

(iii) Given the indices, any two chosen vectors jointly dominate $\mu_{1/2}$ on their $2d$ bits.

(iv) Given the indices and a chosen x, another chosen y is a product distribution with $y_t = 1$ where $x_t = 0$, and $\Pr[y_t = 1] = \frac{1}{2}$ where $x_t = 1$.

Proof. (i), (ii) and (iv) follow from the construction; (ii) and (iv) are products with marginals $\geq \frac{1}{2}$. For (iii), couple one coordinate. The pair is uniform on $\{(1,1),(1,0),(0,1)\}$. Map the points $(1,1),(1,0),(0,1)$ of $\mu_{1/2}^{\otimes 2}$ to themselves, and split $(0,0)$ into three masses of $\frac{1}{12}$, one to each point. □

Lemma 4 ([3, Lemma 22])**.** *For monotone B and $0 < p < q < 1$, $\Pr_{\mu_q}[B = 0] \leq \Pr_{\mu_p}[B = 0]^{\lfloor q/p \rfloor}$. Hence, with $t := \lfloor 1/(2p) \rfloor$, every ν dominating $\mu_{1/2}$ satisfies $\Pr_\nu[B = 0] \leq \Pr_{\mu_p}[B = 0]^t$.*

2.3 Approximators and star plucking

A *slot* is a set J of one or two vectors from distinct lists. A *conjunct on* J is a monotone function B_J of the bits of J that is *saturated*: $B_J = 1$ whenever some vector of J equals 1^d. Property **(P)** means $\Pr_{\mathcal{D}_1}[B_J = 0] \geq \delta$. Under $\mathcal{D}_0$, a saturated conjunct can be 0 only if all of its vectors are chosen. In particular, $B_J \vee C_{J'} \equiv 1$ on $\mathcal{D}_0$ whenever $J \cup J'$ repeats a list.

Definition 5. An *approximator* is a constant, or $F = \bigwedge_{J \in \mathcal{J}} B_J$ (one conjunct per slot) with a set T of fewer than $2M$ *cores*, such that:

(A1) every conjunct satisfies (P);

(A2) every slot meets T, and each 2-slot J is assigned a core in $J \cap T$. The 2-slots assigned to u are the *petals* of u; their other vectors are distinct.

(A3) *(plucked)* for a petal $J = \{u, x\}$ let $\beta_J(w) := \Pr_{x \sim \mu_p}[B_J(w, x) = 0]$, which is nonincreasing in w. Then $\beta_J(w) \in \{0\} \cup (\eta, 1]$ for all w.

(A4) with $Q_u(w) := [\#\{J : \beta_J(w) > \eta\} < \theta/\eta]$ (monotone, $Q_u(1^d) = 1$): either $B_{\{u\}} \leq Q_u$ (*enforced*) or $\Pr_{\mu_p}[Q_u = 0] < \delta$ (*virtual*).

Normalization $N(G)$ of a conjunction G of saturated conjuncts:

1. Drop conjuncts violating (P).
2. Take a maximal family $\mathcal{M}$ of conjuncts with disjoint slots. If $|\mathcal{M}| \geq M$, output 0. Otherwise set $T = \bigcup \mathcal{M}$ and assign cores.
3. *Pluck*: replace each petal B_J by $B_J \vee [\beta_J(u) \leq \eta]$. Then, if $\Pr_{\mu_p}[Q_u = 0] \geq \delta$, AND Q_u into $B_{\{u\}}$; otherwise Q_u is virtual.
4. Drop petals that now violate (P). This only raises Q_u, so (A4) is preserved.

Gates.

- Leaves are their own approximators.
- Constant children are handled exactly: $\mathrm{AND}(0, \cdot) = 0$, $\mathrm{AND}(1, F) = F$, $\mathrm{OR}(1, \cdot) = 1$, $\mathrm{OR}(0, F) = F$.
- AND: $N(F_g \wedge F_h)$, merging equal slots.
- OR: $N(G)$ with $G = \bigwedge (B_J \vee C_{J'})$ over the pairs for which $J \cup J'$ is a slot, so $G \geq F_g \vee F_h$.

2.4 Error analysis

At a gate with output A, the $\mathcal{D}_0$-error is $[A = 1, F_g \circ F_h = 0]$ and the $\mathcal{D}_1$-error is $[A = 0, F_g \circ F_h = 1]$.

Lemma 6 (normalization)**.** *If $M \le n$ and G has $\le 4M$ 1-slots and $\le 8Mn + 4M^2$ 2-slots (true at every gate), then $\Pr_{\mathcal{D}_0}[N(G) = 1, G = 0] \le (40M\delta^t + 4M\eta^t)/n$ and $\Pr_{\mathcal{D}_1}[N(G) = 0, G = 1] \le (1-\delta)^M + 2Me^{-\theta}$.*

Proof. Drops. A dropped conjunct is 0 on $\mathcal{D}_0$ only if its slot is chosen (probability $n^{-|J|}$), and then with probability $\le \delta^t$ (Lemmas 3(ii)/(iii) and 4). In total this is $\le 20M\delta^t/n$.

Switch-off. A change needs u, x chosen and $B_J(u, x) = 0$ with $\beta_J(u) \le \eta$. Given u this has probability $\le \eta^t$ (Lemmas 3(iv) and 4). Over fewer than $2M$ cores with $\le 2n$ petals each, this is $\le 4M\eta^t/n$.

Truncation. The members of $\mathcal{M}$ are independent under $\mathcal{D}_1$, each failing with probability $\ge \delta$.

Enforcing Q_u. Given u with $Q_u(u) = 0$, at least θ/η petals with distinct, independent partner vectors each fail with probability $> \eta$. So all of them equal 1 with probability $\le e^{-\theta}$. □

Lemma 7 (OR gate)**.** $\Pr_{\mathcal{D}_0}[G = 1, F_g = F_h = 0] \le 4M\delta^t/n + 26M^2(\theta/\eta)^2/n^3$.

Proof. Let $\mathcal{V}$ be the event that some chosen core with a *virtual* Q_u has $Q_u(u) = 0$. Then $\Pr[\mathcal{V}] \le 4M\delta^t/n$. Work outside $\mathcal{V}$, on a sample with $G = 1$ and $F_g = F_h = 0$.

Failing conjuncts lie inside the chosen triple C. If failing J (of F_g) and J' (of F_h) had $J \cup J' \subsetneq C$, then $B_J \vee C_{J'}$ would be a failing conjunct of G. So every failing pair spans C.

If a side is a petal of a core u with $Q_u(u) = 0$, then Q_u is enforced, so $B_{\{u\}}$ fails too, and replacing that side by $\{u\}$ yields another failing pair, which must also span. Two single vectors cannot span, so we end with either (a) a single vector against a petal whose core passes Q, or (b) two petals whose cores pass Q.

A failing petal whose core passes Q is moderate ($\beta > \eta$), and fewer than θ/η petals per core value are moderate. By Lemma 3(i), the chosen vector of another list is one of them with probability $< \theta/(\eta n)$.

Two spanning 2-slots share exactly one vector. Counting the configurations:

- (a): $\le 2(2M)^2$ configurations, each with probability $\le n^{-2}\theta/(\eta n)$;
- (b), same core on both sides: $< 2M$ configurations, each $\le n^{-1}(\theta/(\eta n))^2$;
- (b), the core of one side is a petal of the other: $\le 2(2M)^2$, each $\le n^{-2}\theta/(\eta n)$;
- (b), a common petal: $\le (2M)^2$, each $\le n^{-2}\theta/(\eta n)$.

The total is $\le 26M^2(\theta/\eta)^2/n^3$. □

Proof of Theorem 1. Let the circuit C have s gates and output approximator F. By induction, $F \ge C$ outside the union of the $\mathcal{D}_1$-errors, and $F \le C$ outside the union of the $\mathcal{D}_0$-errors.

- If $F \equiv 0$, then $\frac{1}{2} \le s[(1-\delta)^M + 2Me^{-\theta}]$.
- Otherwise every conjunct meets T, so $\Pr_{\mathcal{D}_0}[F = 0] < 2M/n \le \frac{1}{2}$. Hence $\frac{1}{2} \le s\varepsilon_0$ with $\varepsilon_0 = (44M\delta^t + 4M\eta^t)/n + 26M^2(\theta/\eta)^2/n^3$.

Take $t = \lceil 64/\varepsilon \rceil$, $p = 1/(2t)$, $M = n^{\varepsilon/8}$, $\theta = n^{\varepsilon/16}$, $\delta = n^{-\varepsilon/16}$ and $\eta = n^{-4/t}$. Then $\delta^t, \eta^t \le n^{-4}$ and $\theta/\eta \le n^{\varepsilon/8}$, so $\varepsilon_0 \le 27n^{-3+\varepsilon/2}$ and $(1-\delta)^M \le e^{-n^{\varepsilon/16}}$. Therefore $s \ge \min\{e^{n^{\varepsilon/16}}/(6n^{\varepsilon/8}), n^{3-\varepsilon/2}/54\} \ge n^{3-\varepsilon}$ for large n, with $c = 4(2t)^3 = O(\varepsilon^{-3})$. □

Remark. For two lists every 2-slot spans, only 1-slots survive, and the argument reduces essentially to [3]. (One difference: truncation here uses a maximal disjoint family and also runs at OR gates.) For $k \ge 4$, conjuncts read up to $k-1$ vectors, and plucking would be needed at cores of every size up to $k-2$. This is left open.

3 Limits of routes to refuting OVH

3.1 Block-CNF collapses for constant block size

A *block-CNF* has variables $x_\ell \in [D]$ ($\ell \le k$) and clauses that are disjunctions of block-literals $[x_\ell \in T]$. Let $n = k \log_2 D$. k-OV with D vectors per list in dimension d is a block-CNF with d clauses. We assume (H): #q-SAT is in $2^{\delta n}$ time for all q and $\delta > 0$, as claimed by [1, 2].

Theorem 8. *Under (H), for fixed D, Δ and $\delta > 0$, block-CNFs with $m \le \Delta n$ clauses can be counted in $O(2^{\delta n})$ time. In particular, k-OV with constant D in dimension $\Delta k \log_2 D$ takes $D^{o(k)}$ time.*

Proof sketch. Branch on any clause wider than w. Either a fixed w-subset C' holds, or every literal of C' fails, which shrinks w domains. With the measure $\sum_\ell \log_2 |R_\ell|$ the number of leaves is $2^{n\eta(w)}$, where $\eta(w) \to 0$. At the leaves, the remaining width-$\le w$ block-CNF is a $(w\lceil \log_2 D \rceil)$-CNF with $O(mD^w)$ clauses, which (H) counts in $2^{\delta' n}$ time. □

With the explicit bound of [2] this extends to $D \approx (\log n)^{1/4}$. Any CNF-expansion method is limited to $D \le \mathrm{polylog} N$, far below OV's $D = \sqrt{N}$. Combined with [7, Lemma 12], ¬ETH removes all SAT-based support for OVH in low dimension.

3.2 Linear deformation designs

Proposition 9. *Formalize the deformation/Taylor-interpolation framework of [1] for #OV $= \sum_{i,j} \prod_o (1 - \alpha_o \beta_o)$ with linear measurements of rank $\le r$ and deformation degree R over Q points. Then for $d = c \log_2 N$ with $c > 2 + 2/\log_2 3 \approx 3.26$, every such design costs $N^{2+\Omega(1)}$. If the inclusion–exclusion of [1] ranges over the $2d$ input variables rather than the d coordinates, $c > 1 + 2/\log_2 3 \approx 2.26$ suffices.*

Proof sketch. A corank argument shows that at each point at least $d - 2r$ input variables move, hence at least $(d - 2r)/2$ coordinates. A determinant argument ($(T - t)^\kappa$ divides $\det \Pi(T)$, whose degree is $\le dR$) gives $Q(d - 2r) \le 2dR$, which forces Taylor order $q \ge d - 2r - 1$. Subquadratic design sums need $r \le 0.631 \log_2 N$, so inclusion–exclusion over coordinates enumerates $\ge 2^{(d-2r)/2-1}$ subsets, each costing $\ge N$: in total $N^{1+(c-1.262)/2-o(1)}$. Over input variables the count is $\ge 2^{d-2r-1}$, giving $N^{c-0.262-o(1)}$. □

3.3 Equality rank of disjointness matrices

Williams [5] showed that if DISJ_k (with entries $[x \cap y = \emptyset]$) has equality rank $2^{o(k)}$, then OVH fails. The equality rank $r(k)$ is the least r with $\mathrm{DISJ}_k = \sum_{i \le r} \alpha_i E_i$, where the E_i are equality matrices $[u(i) = v(j)]$. Earlier computer searches used only 0/1 coefficients. For fixed r, feasibility depends only on the *type* of α (which subset sums equal 0 or 1). We enumerate the finitely many realizable types exactly and solve one SAT instance per type (`code/equality_rank_sat.py`).

Theorem 10 (computer-assisted)**.** *Over $\mathbb{R}$: $r(1) = 2$, $r(2) = 2$, $r(3) = 3$, $r(4) = 4$. For $k = 4$, $r = 3$ there are 32 types: 8 are trivially infeasible and 24 are UNSAT.*

Proposition 11 (computer-assisted)**.** $4 \le r(5) \le 5$. *Of the 168 types for $r = 4$, 20 are trivial and 147 are UNSAT. The remaining type is a pure partition. So $r(5) = 4$ iff the ones of* DISJ_5 *split into 4 equality matrices. This is* open*: it was not decided by several hours of CDCL search, including a cube-and-conquer run in which all 24 cubes hit a 900 s limit.*

Proposition 12. *The fractional cover number of* DISJ_k *by equality submatrices is* $7/4$, $23/10$, ≈ 2.692 *and* ≈ 3.306 *for $k = 2, 3, 4, 5$ (certified lower bounds $7/4$, $23/10$, $13471/5005$, $8255/2499$).*

Weights of either sign, valid for partitions, give the same values. So weighting arguments cannot prove $r(4) \geq 4$ or $r(5) \geq 5$, even for partitions. The computation is in `code/fractional_cover.py`.

Beating the $\tilde{O}(n \cdot 1.26^d)$ #OV bound of [6] with a constant-size decomposition needs $r(k) < 1.26^k$. If $r(5) = 5$, the first chance is $r(7) = 5$.

4 Verification status

Result	Verification so far
Theorem 1, Corollary 2	Full proof; 3 AI referee reads, minor fixes applied. **No human or formal check.**
Theorem 8, Proposition 9	Proof sketches with numerical sanity checks; not reviewed. Theorem 8 is conditional on (H), i.e. on [1, 2].
Theorem 10, Proposition 11	CaDiCaL 1.9.5, logs in `data/`. No DRAT certificates; type enumeration not independently checked.
Proposition 12	Floating-point LP; lower bounds have exact certificates, optimality does not.

5 Open problems

1. Human verification of Theorem 1, especially Lemma 7 and (A3)–(A4); ideally also a Lean formalization.
2. Monotone k-OV at $d = O(\log n)$ for all $k \geq 4$, via recursive star plucking.
3. The sharp monotone trade-off for two lists: is any monotone circuit subquadratic at some $c > 1$?
4. Decide $r(5)$, e.g. by lookahead cubing with DRAT proofs.

Use of AI

The author chose the problem and directed the sessions; Claude (Anthropic, Claude Opus 5.5) did the literature search, proofs, experiments, code and write-up. Review was by separate instances of the same model.

References

[1] OpenAI. *The Exponential Time Hypothesis is False.* Preprint, 25 Oct 2026, github.com/openai/math.

[2] Anthropic. *A $2^{o(n)}$ Time Algorithm for 3-SAT.* Lean-formalized preprint, Oct 2026, github.com/anthropics/formal-math.

[3] T. Choudhury, N. Limaye, K. Sreenivasaiah, S. Srinivasan. *New and Improved Concrete Lower Bounds for Orthogonal Vectors.* arXiv:2607.23799, 2026.

[4] R. Williams. *A new algorithm for optimal 2-constraint satisfaction and its implications.* TCS, 2005.

[5] R. Williams. *The Orthogonal Vectors Conjecture and Non-Uniform Circuit Lower Bounds.* FOCS 2024.

[6] Alman and Li. *Kronecker Powers, Orthogonal Vectors, and the Asymptotic Spectrum.* arXiv:2509.14489, 2025.

[7] C. Calabro, R. Impagliazzo, R. Paturi. *A duality between clause width and clause density for SAT.* CCC 2006.

[8] A. Razborov. *Lower bounds on the monotone complexity of some Boolean functions.* 1985.

[9] N. Alon, R. Boppana. *The monotone circuit complexity of Boolean functions.* Combinatorica, 1987.

[10] É. Tardos. *The gap between monotone and non-monotone circuit complexity is exponential.* Combinatorica, 1988.